



Fig. 3: Ngspice simulation output

How to use Ngspice

- Using MS Explorer window (interactive mode):
 - Double-click on ngspice.exe in C:\Spice\bin
 - In the Ngspice window, type the following:
 - cd c:\spice\tests\bsim4
 - source comprt.cir
 - run
 - display
 - plot v(1) v(2) v(3) v(8) v(9)
 - write comprt2.raw
 - quit
 - Using MSYS window (interactive mode):
 - Open an MSYS window
 - In the MSYS window, type the following codes:
 - export PATH=\$PATH:/c/Spice/bin
 - cd /c/Spice/tests/bsim4
 - ngspice comprt.cir
 - Ngspice window will open up; in that window, type:
 - run
 - display
 - plot all
 - Write the data into a raw file to the actual directory /c/Spice/tests/bsim4 using the following command:
 - write comprt.raw
 - and then type "quit"
- Note.** MSYS is specific about the file name and directory. Also, you should have administrator rights to the directory you are running these files from.
- If the circuit file you are using is in a different directory (for example, C:\test\comprt.cir) than the interactive mode, then the following codes can be used to call Ngspice:
 - ngspice C:\test\comprt.cir or
 - ngspice C:\test\comprt.cir

over other circuit simulators.

Built-in models. For every semiconductor device, Ngspice has a built-in model that only requires a

few specifications to be set. In case of bipolar junction transistor (BJT), Ngspice has three built-in models, all of which are based on the integral charge model of Gummel and Poon; BJT reduces it to Ebers-Moll model, which is the simplest one.

Event-driven simulation. User-defined nodes allow event-driven simulation, where users are allowed to specify the nodes for data propagation of values other than current, voltage and digital states. However, state values in these cases might be an arbitrary data type. Ngspice implements these user-defined nodes in such a way that the data structure defines only the digital state held by a strength value and a Boolean logic state.

Supports multiple analyses. Ngspice supports multiple analyses like AC/DC analysis, transient analysis, pole-

zero analysis, small-signal-distortion analysis, sensitivity analysis and noise analysis, where most of these are exclusively used by analogue

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